# Andrew Salem

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Canadian citizen (TN-eligible for US roles) | Open to relocation

Machine learning engineer with a life sciences foundation. I train generative models (flow matching, diffusion, fine-tuned language models) and own the data pipelines, distributed training and evaluation behind them, across datasets from a 124-million-item corpus to regulated analytical records. Bioengineering background spanning protein and molecular design, medical imaging and wet-lab synthetic biology.

## Education

**MSc, Life Sciences Engineering** | ***GPA 5.73/6.0*** — Sep. 2023 – Feb. 2026
EPFL, Swiss Federal Institute of Technology Lausanne — Lausanne, Switzerland

**BEng, Bioengineering** | ***GPA 4.0/4.0, Charles Michael Morssen Gold Medal*** — Aug. 2019 – May 2023
McGill University — Montreal, QC, Canada

## Experience

**Machine Learning Engineer (contract)** — Mar. 2026 – Oct. 2026
EPFL, Dal Peraro Lab — Lausanne, Switzerland

- Trained generative models (flow matching, diffusion) on 3D point-cloud and graph data in PyTorch and PyTorch Geometric, including equivariant architectures and custom geometric loss terms.
- Owned the end-to-end data platform on an HPC cluster, from ingestion and validation through to sharded datasets for distributed training.
- Fine-tuned transformer language models with RL and direct preference optimization, using an automated scorer as the reward; gains verified by significance testing.
- Diagnosed a train/inference distribution mismatch costing most of the model's structural accuracy, and recovered it through a sampler change alone.
- Standardised experiment tracking, checkpointing and resumable runs, keeping results comparable across weeks of cluster time.
- Set technical direction for two junior researchers.

**Data Quality Analyst (intern)** — Sep. 2024 – Feb. 2025
Merck KGaA — Aubonne, Switzerland

- Validated 49 automated analytical data pipelines through a global migration, in a regulated (GxP) environment.
- Built performance dashboards in SAP, Palantir Foundry and TrackWise to track project lifecycles and quantify efficiency gains.
- Worked across analytical, IT and quality teams to sign off each migrated method against its specification.

**Machine Learning Consultant** — May 2023 – Aug. 2023
CoreSlicer — Montreal, QC

- Built an end-to-end computer-vision pipeline (PyTorch, OpenCV) for 3D volumetric segmentation from DICOM data, reaching a Dice coefficient of 0.87.
- Handled the full lifecycle: preprocessing and augmentation of voxel data, model training and evaluation, then deployment on AWS as a scalable inference service.

**Research Assistant, Scientific Computing** — Sep. 2021 – Aug. 2022
McGill University — Montreal, QC

- Automated acquisition and analysis of high-throughput experimental data, replacing manual measurement with scripted image-processing pipelines.
- Maintained shared laboratory instrumentation and calibration procedures used across several concurrent projects.

**Backend Developer (apprentice)** — May 2020 – Sep. 2020
Axiom SSC — Remote

- Refactored SQL queries to cut data-retrieval latency and resolved server-side defects on Microsoft Azure, raising service uptime to 99.7%.

## Selected Projects

**Foundation Model for Molecular Representation Learning** — Mar. 2026 – present

- Assembled and cleaned a 124-million-item dataset, with the full featurization and quality pipeline.
- Pretrained a hierarchical graph encoder with self-supervised reconstruction objectives (90% weighted node/edge accuracy); frozen embeddings recover downstream regression targets at $R^2 \approx 0.9$ under linear probing.

**Ligase: Benchmarking Harness for Language Models and GNNs** — Jul. 2026 – Oct. 2026

- Built a reproducible evaluation framework comparing transformer embeddings against graph neural networks across 18 ablations; showed frozen encoders with $\leq$1M-parameter heads match or beat far larger models.
- Eliminated a 40 GB transient memory spike in batched inference with token-budgeted dynamic batching, and added self-calibrating runtime test gates to catch silent metric drift.
- Designed a leakage-free benchmark over 131k comparisons, backed by CI/CD and invariance property tests.

**GPTeeHee: LLM Fine-Tuning and Quantized Deployment** — Spring 2025

- Fine-tuned the Qwen3 language model on domain data with a custom quantization setup for local inference, reaching 51% on MMLU.

**Gynia: Retrieval-Augmented Generation Application** — Spring 2025

- Built a full-stack RAG system (OpenAI API, PostgreSQL, Streamlit) with multi-agent retrieval and automated report generation.

## Research

**Master's Thesis: Generative Models for Structured Scientific Data** Aug. 2025 – Feb. 2026
EPFL (Thesis Award nomination) Lausanne, Switzerland

- Designed a graph variational autoencoder with bi-directional cross-attention and a permutation-invariant set transformer, trained on a multi-task objective; 92% top-level classification accuracy and 0.87 AUROC.
- Built and compared two generative decoders (frame-based vs. all-atom flow matching, ∼340M parameters each) over 500k training steps, and showed that decomposing the task outperforms a single end-to-end model at this data scale.
- Designed the evaluation: leakage-free clustered splits, per-choice ablations, and latent analysis recovering a known hierarchy unsupervised.

**Research Intern** Feb. 2025 – Aug. 2025
Radenovic Lab, EPFL Lausanne, Switzerland

- Assembled a custom interferometric scattering (iSCAT) microscope from optical components for label-free measurements in biological systems.
- Wrote a MATLAB simulation of fluorophore absorption and emission dipoles.

**Research Intern** Jan. 2024 – Aug. 2024
Gönczy Lab, EPFL Lausanne, Switzerland

- Built an automated image-analysis pipeline (Random Forest segmentation and a feed-forward network for particle assignment) to track mitochondria in *C. elegans* embryos.
- Maintained transgenic *C. elegans* strains and imaged early embryogenesis by DIC and spinning-disk confocal microscopy.

**Research Apprentice** Jan. 2023 – Aug. 2023
Kinsella Lab, McGill University Montreal, QC

- Developed a decellularized ECM–collagen hydrogel for 3D organoid culture of pancreatic adenocarcinoma.
- Measured the cytotoxicity of stem-cell-derived natural killer (NK) cells against pancreatic cancer models using flow cytometry, western blotting and microscopy.

**Research Apprentice, Synthetic Biology** Sep. 2022 – May 2023
Ignea Lab, McGill University Montreal, QC

- Engineered *S. cerevisiae* to biosynthesize momilactone through heterologous expression of plant genes; optimized shake-flask fermentation as the basis for a 50 L+ bioreactor scale-up.
- Designed and built expression plasmids (SnapGene, Gibson Assembly), validated by PCR, sequencing and GC-MS metabolite quantification.

**Research Apprentice** May 2021 – Aug. 2022
Nicolau Lab, McGill University Montreal, QC

- Built a Langevin-equation model of actin motility in MATLAB, using Runge–Kutta methods to speed up simulations.
- Fabricated PDMS casting masters by photo-resin 3D printing and SU-8 photolithography in an ISO 4 cleanroom; scripted vascular branching models in Fusion 360.

## Skills

**Machine learning**: Generative models (flow matching, diffusion, VAEs), transformers and LLM fine-tuning (LoRA, RL, DPO), graph neural networks, self-supervised pretraining, computer vision, Bayesian optimization, RAG
**Languages and frameworks**: Python, PyTorch, PyTorch Geometric, NumPy, scikit-learn, OpenCV, SQL, MATLAB, Java
**Infrastructure**: HPC and distributed training, Docker, Git, CI/CD, AWS (SageMaker), Microsoft Azure, PostgreSQL, Streamlit
**Laboratory**: Yeast metabolic engineering, Gibson Assembly, PCR, GC-MS, cell culture (3D organoids), flow cytometry, confocal and DIC microscopy, microfabrication (SU-8, ISO 4 cleanroom)
**Quality and regulatory**: GxP/GMP, data-integrity validation, SAP, Palantir Foundry, TrackWise
**Languages**: English and French (native), Arabic (fluent), German (B2), Spanish (B1)

## Honors

EPFL Master's Thesis Award nomination (2026) | Swiss Excellence Fellowship (2023–2025) | Charles Michael Morssen Gold Medal, McGill (2023) | Lieutenant Governor's Bronze Medal (2023) | NSERC USRA (2021, 2022)

## Publication

Kang'iri, S. M., **Salem, A.**, Nicolau, D. V., & Nitta, T. (2022). Effects of defective motors on the active transport in biosensors powered by biomolecular motors. *Biosensors and Bioelectronics*, 203, 114011. doi.org/10.1016/j.bios.2022.114011

## Leadership and Community

**Head of Logistics and Operations**, SV Industry, EPFL (2024–2026) | **Lead STEM Tutor**, Jeunesse School, Lausanne (2024–2026) | **First-Year Master's Representative**, EPFL (2023–2024) | **Volunteer**: Red Cross of Vaud, Hope&Cope, Les AmiEs du boisé Dora-Wasserman (2019–present)